Section 4: Elementary Statistical Inference

In a statistical problem with a random variable X, we have something of interest, but its distribution may often be unknown or, when $f(x)$ or $p(x)$ is known then θ, a parameter of interest, is unknown. For example,

$$X \sim Exp(\theta), \quad \text{where } \theta = \lambda \text{ is the unknown parameters}$$

$$X \sim bin(n, \theta), \quad \text{where } \theta = p \text{ is the unknown parameters}$$

$$X \sim N(\theta), \quad \text{where } \theta = \begin{bmatrix} \mu \\ \sigma \end{bmatrix} \text{ is a vector of unknown parameters}$$

Thus, we now update our distribution for X through the PDF or PMF by:

$$X = \begin{cases} f(x;\ \theta), & X \sim \text{cont RV} \\ p(x;\ \theta), & X \sim \text{disc RV} \end{cases}$$

where $\theta \in \Omega$ for a specific set, e.g. for $X \sim Exp(\theta)$, the set $\Omega = \{\theta \mid \theta > 0\}$ and θ is our estimate of λ.

In either case of an unknown distribution or parameter, our information and inferences comes from our sample of X.

Notation: $\underbrace{X_1, X_2, \cdots, X_n}_{\text{uppercase}}$ are RVs of "observed" values from X

$x_1, x_2, \cdots, \underbrace{x_n}_{\text{lowercase}}$ are "real" values of X

Definition 4.1.1: If the random variables $X_1, X_2, \dots, X_n$ are independent and identically distributed (iid), the random variables constitute a **random sample** of size n with a common distribution, i.e. $X \sim f(x)$ or $p(x)$.

Definition 4.1.2: Let $X_1, X_2, \dots, X_n$ be a random sample of random variables and $T = T(x_1, x_2, \dots, x_n)$ be a function of the sample, then T is called a **statistic**.

Once a sample is drawn from X, t is the **realization** of T, where $t = T(x_1, x_2, \dots, x_n)$. For example, let $x = \{1, 2, \dots, 5\}$, where $X \sim f(x)$ and $\exists\ E(X)$, then if we define $T = \frac{1}{n}\sum x_i$, if $t = T(1, 2, \dots, 5) = 3$, then t is our sample mean $\bar{x}$, i.e. our statistic.

Thus, we consider a statistic T the **point estimator of $\boldsymbol{\theta}$** and t, the realization, is an estimate of θ based on our sample n, e.g. $\bar{x}$, our statistic and sample mean, is an estimate of μ, our parameter of interest.

Definition 4.1.3: Let $X_1, X_2, \dots, X_n$ be a sample from a random variable with PDF $f(x; \theta)$, $\theta \in \Omega$. Let $T = T(x_1, x_2, \dots, x_n)$ be a statistic, if $E(T) = \theta$ then T is an **unbiased** estimator of θ.

Example 4.1: let $x_1, x_2, \dots x_n \overset{iid}{\sim} N(\mu, \sigma)$ and $T = \frac{1}{n}\sum x_i$

then $E(T) = E\left(\frac{1}{n}\sum x_i\right) = \frac{1}{n}E(X_1 + \dots + X_n)$

$= \frac{1}{n}\left[E(X_1) + \dots + E(X_n)\right]$ X_i are ind.

$= \frac{1}{n}\left[\mu + \dots + \mu\right]$ x's are actually $N(\mu, \sigma)$

$= \frac{1}{n}(n \cdot \mu)$

$= \mu \Rightarrow T$ is unbiased estimator of μ

In general, to find these estimators, we first look at the information in the sample and the parameter θ, as well as the joint distribution. Since we **assume that $\boldsymbol{X_i, X_j}$ are independent** $\forall\ \boldsymbol{i, j}, \ni \boldsymbol{i \neq j}$, as well as being identically distributed, then the joint distribution $f(x_1, x_2, \dots, x_n;\ \theta)$ is defined as follows:

$$f(x_1, x_2, \dots, x_n; \theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \dots \cdot f(x_n; \theta)$$
$$= \prod_{i=1}^{n} f(x_i; \theta)$$
$$= L(\theta) = L(\theta; x_1, x_2, \dots, x_n)$$

where $L(\theta)$ is referred to as the **likelihood function** of the random sample and if the estimate of θ maximizes $L(\theta)$ and is unique, we refer to that estimate as the **maximum likelihood estimator (MLE)**, where $\hat{\theta} = \text{Argmax}\, L(\theta)$

↳ argument of the maxima, which are the points on the domain of some function at which the function values are maximized

Although we can work with the likelihood function, we can take advantage of the fact that the logarithm of the same function is easier to work with (mathematically) while the maxima will work for both. It is important to note that the log of the likelihood, where $l(\theta) = \log L(\theta)$ references the natural logarithm, i.e. ln, and not log-base 10. Therefore, to find the maxima of $l(\theta)$, we need to determine the θ such that,

$$\frac{\partial}{\partial \theta} l(\theta) = 0$$

and if θ is a vector, then we'll have to solve a system of equations.

Example 4.2:

let $x_1, x_2, \dots, x_n \overset{iid}{\sim} \Gamma(1, \theta)$, then

$x_i \sim f(x_i; \theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}}$

let $\theta = \frac{1}{\lambda} \Rightarrow x_i \sim f(x_i; \theta) = \lambda e^{-\lambda x}$

$\Rightarrow x_i \sim Exp(\lambda = \frac{1}{\theta})$

$l(\theta) = \ln(L(\theta)) = \ln\left(\prod_{i=1}^{n} f(x_i; \theta)\right)$

$= \ln\left(\prod_{i=1}^{n} \lambda e^{-\lambda x}\right)$

$= \ln\left(\lambda e^{-\lambda x_1} \cdot \dots \lambda e^{-\lambda x_n}\right)$

$$= \sum_{i=1}^{n} \ln\left(\lambda e^{-\lambda x_i}\right)$$

$$= \sum_{i=1}^{n} \ln(\lambda) + \ln\left(e^{-\lambda x_i}\right)$$

$$= n\ln(\lambda) + \sum_{i=1}^{n} (-\lambda x_i)$$

$$= n\ln(\lambda) - \lambda \cdot \sum_{i=1}^{n} x_i$$

$$\Longrightarrow \frac{\partial}{\partial\theta} \ell(\theta) = \frac{\partial}{\partial\theta}\left[n\ln(\lambda) - \lambda \cdot \sum_{i=1}^{n} x_i\right]$$

$$= \frac{\partial}{\partial\lambda}\left[n\ln(\lambda) - \lambda \cdot \sum_{i=1}^{n} x_i\right]$$

$$= n \cdot \frac{1}{\lambda} - \sum_{i=1}^{n} x_i = 0$$

$$\Longrightarrow \frac{n}{\lambda} = \sum_{i=1}^{n} x_i$$

$$\frac{1}{\lambda} = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x}$$

"hat" is for the estimator

$$\therefore \underbrace{\hat{\theta}}_{MLE} = \frac{1}{\hat{\lambda}} = \frac{1}{n}\sum x_i = \bar{x}$$

$$E(\bar{X}) = \mu = \theta = \frac{1}{\lambda} \quad \therefore \text{ unbiased estimator}$$

As we can see, even though $X \sim Exp\left(\lambda = \frac{1}{\theta}\right)$, the **sample mean is still an unbiased estimator and the MLE**. Moreover, we can see (mathematically), the importance of the independence assumption in statistics, as the likelihood function is much more difficult to work with/calculate if the assumption were to be violated.

Example 4.3: $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma)$

$$\ell(\theta) = \ln(L(\theta)) = \ln\left(\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x_i-\mu}{\sigma}\right)^2}\right)$$

$$= \ldots$$

$$= -\frac{n}{2}\ln(2\pi) - n\ln(\sigma) - \frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)^2$$

$$\Rightarrow \frac{\partial}{\partial\mu}\ell(\theta) = -\frac{2}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)\cdot\left(-\frac{1}{\sigma}\right)$$

$$= \frac{1}{\sigma}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)$$

$$= \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0$$

$$=$$

$$\Rightarrow \frac{\partial}{\partial\sigma}\ell(\theta) = -\frac{n}{\sigma} - \frac{2}{2}\sum_{i=1}^{n}\left(\frac{x_i-\mu}{\sigma}\right)\cdot\left(-\frac{x_i-\mu}{\sigma^2}\right)$$

$$= -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

Since σ^2 is a function based on μ, we need to solve these two together to get the MLE $\hat{\theta}$

$$\frac{1}{\sigma^2}\sum(x_i-\mu) = 0 \Rightarrow \hat{\mu} = \bar{x} = \frac{\sum x_i}{n}$$

$$-\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\sum_{i=1}^{n}(x_i-\mu)^2 = n\sigma^2$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2}$$

$$= \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

recall: $s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2$

$\sigma^2 = \frac{1}{n}\sum(x_i-\bar{x})^2$

Similar to the exponential example, the statistic $\hat{\mu} = \bar{x}$ is the MLE and it is an unbiased estimator of μ; however, when we consider $\hat{\sigma}^2$ (or simply s^2) we get a different story,

$$
\begin{aligned}
E(\hat{\sigma}^2) &= E\left(\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2\right) \\
&= E\left(\frac{1}{n}\left[\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right]\right) \\
&= E\left(\frac{1}{n}\left[\sum x_i^2 - n\bar{x}^2\right]\right) \\
&= \frac{1}{n}E\left(\sum x_i^2 - n\bar{x}^2\right) \\
&= \frac{1}{n}\left[E\left(\sum x_i^2\right) - nE(\bar{x}^2)\right] \\
&= \frac{1}{n}\left[\sum E(x_i^2) - nE(\bar{x}^2)\right] \\
&= E(x_i^2) - E(\bar{x}^2) \\
&= \left[\sigma^2+\mu^2\right] - \left[\sigma_{\bar{x}}^2 + \mu_{\bar{x}}^2\right] \\
&= \sigma^2 + \mu^2 - \sigma_{\bar{x}}^2 - \mu^2 \\
&= \sigma^2 - \sigma_{\bar{x}}^2
\end{aligned}
$$

Since $x_i \overset{iid}{\sim} N(\mu,\sigma)$, all must have the same second moment

CLT: $\bar{x} \sim N(\mu_{\bar{x}} = \mu,\ \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}})$

$$
\begin{aligned}
E(\hat{\sigma}^2) &= \sigma^2 - Var(\bar{x}) \\
&= \sigma^2 - var\left(\frac{1}{n}\sum x_i\right) \\
&= \sigma^2 - \frac{\sigma^2}{n} \\
&= \left(\frac{n-1}{n}\right)\sigma^2
\end{aligned}
$$

$\left(\frac{1}{n}\right)^2 Var\left(\sum x_i\right) = \frac{1}{n} Var(x) = \frac{\sigma^2}{n}$

Thus, $\hat{\sigma}^2$ is a MLE but since $E(\hat{\sigma}^2) \neq \sigma^2$, it is **not an unbiased estimator** of σ^2; however, it is **asymptomatically unbiased**; that is,

$$\lim_{n\to\infty} \left(\frac{n-1}{n}\right)\sigma^2 = \sigma^2 \cdot \lim_{n\to\infty}\left(\frac{n-1}{n}\right) = \sigma^2$$

So, although not unbiased, as the sample size n increases, our MLE becomes unbiased. The difference in the scaling factor $\left(\frac{1}{n}\right)$ vs. $\left(\frac{1}{n-1}\right)$ for the MLE $\hat{\sigma}^2$ and our "known" sample variance relates to this asymptomatic property, which explains why

$$S^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$$

based on second variance formula above: $s^2 = \frac{1}{n-1}\left[\sum x_i^2 - \frac{(\sum x_i)^2}{n}\right]$

$$E(s^2) = E\left(\frac{1}{n-1}\sum(x_i-\bar{x})^2\right)$$

$$= \frac{1}{n-1}\left[E\left(\sum x_i^2\right) - nE(\bar{x}^2)\right]$$

$$= \frac{n}{n-1}\left[E(x_i^2) - E(\bar{x}^2)\right]$$

$$= \frac{n}{n-1}\left[\left(\frac{n-1}{n}\right)\sigma^2\right] = \sigma^2$$

i.e. S^2 is an unbiased estimator

Example 4.4 (Exercise 4.1.1 via Hogg): Twenty motors were put on test under a high-temperature setting. The lifetimes in hours of the motors under these conditions are given below.

1	4	5	21	22	28	40	42	51	53
58	67	95	124	124	160	202	260	303	363

Suppose we assume that the lifetime of a motor under these conditions, X, has a $\Gamma(1,\theta)$ distribution.

$\hookrightarrow X \sim Exp(\lambda = \frac{1}{\theta})$

(A) Do you think $X \sim \Gamma(1,\theta)$ is reasonable

We'll leave you to decide; however, keep in mind that when we example plots such as these, we can't "confirm" $X \sim \Gamma(1,\theta)$, but rather conclude whether it not unreasonable. This line of thought falls within the same way of thinking when doing a hypothesis test. When we FTR a hypothesis, we are not confirming it, we simply do not have sufficient evidence of the contrary; likewise, we are not confirming that the data came a $\Gamma(\alpha,\beta)$ distribution we simply do not have sufficient evidence to say that it did not come from this distribution.

(B) Obtain the MLE

$X \sim \Gamma(1, \theta) \equiv Exp(\frac{1}{\theta})$

$f(x;\theta) = \frac{1}{\theta} e^{-\frac{x}{\theta}} \equiv \lambda e^{-\lambda x}$, where $\lambda = \frac{1}{\theta}$

$\ell(\theta) = \sum_{i=1}^{n} \ln(f(x,\lambda)) = n\ln(\lambda) - \lambda \sum x_i$

$\Rightarrow \frac{\partial}{\partial\theta} \ell(\theta) = \frac{\partial}{\partial\lambda} \ell(\lambda) = \frac{n}{\lambda} - \sum x_i = 0$

$\Rightarrow \frac{1}{\lambda} = \theta = \bar{x}$

$\Rightarrow \hat{\theta} = \bar{x} = 101.15$

(C) What is the sample median?

sample median $= \hat{\eta} = 55.5$

Since the median is the 50th percentile

$\Rightarrow P_r(x \leq \eta) = 0.50$

$\Rightarrow \int_0^{\eta} \lambda e^{-\lambda} dx = 0.50$

$\Rightarrow e^{-\lambda\eta} = 0.50$

$\Rightarrow \hat{\eta} = \frac{\ln(2)}{\lambda} \Leftrightarrow \hat{\theta}\ln(2)$

(D) Based on the MLE, what is another estimate of the median?

$\hat{\eta} = \hat{\theta} \ln(2) = \overline{X} \cdot \ln(2) = 101.15 \cdot \ln(2) = 70.112$

Q: Which is the better measure of central tendency for the type of data we are working with?

if we have skewed data, the median is the better choice

Now that we've shown how we can estimate θ by a statistic $T = T(X_1, X_2, \dots, X_n)$, which is represented by $t = t(x_1, x_2, \dots, x_n)$ when our sample is drawn, a question of "**How good of an estimate?**" often arises. Although we know that $\bar{x}$ is an unbiased estimate (and MLE) of μ, it is very rare that $\bar{x} = \mu$. Thus, how much did we miss θ by when using $\hat{\theta}$ is an important question to address.

Definition 4.2.1: Let $x_1, x_2, \dots, x_n$ be a sample of random variable X, where $X \sim f(x;\ \theta), \theta \in \Omega$. Let $\exists\ \alpha \ni \{0 < \alpha < 1\}$ be specified, and $L = L(x_1, x_2, \dots, x_n)$ and $U = U(x_1, x_2, \dots, x_n)$ be two statistics. Then we say that the interval (L, U) is a $100(1-\alpha)\%$ **confidence interval** for θ if: $1 - \alpha = P(\theta \in [L, U])$. That is, the probability that (L, U) includes θ is $(1-\alpha)$ or $100(1-\alpha)\%$.

Continuing our trend of realization, l and u represent the realized values of a confidence interval once the sample is drawn. **Importantly, either $(\boldsymbol{l}, \boldsymbol{u})$ contains $\boldsymbol{\theta}$ or it does not. If we were to repeat the sample over and over gain (say 100 times), then we would expect $\boldsymbol{\theta}$ to fall within those 100 $(\boldsymbol{l}, \boldsymbol{u})$ intervals $\mathbf{100(1-\alpha)}$ times and would fall outside those 100 $(\boldsymbol{l}, \boldsymbol{u})$ intervals $\mathbf{100\alpha}$ times**.

Let's say we have 2 such random samples. Although we may not know if θ falls in either (l_1, u_1) or (l_2, u_2), we can still compare the two intervals. Once such comparison is the **efficiency**, where (L_1, U_1) is more efficient than (L_2, U_2) if $E(U_1 - L_1) \leq E(U_2 - L_2)\ \forall\ \theta \in \Omega$; that is, we are literally comparing the lengths of the intervals. This should make sense when we consider what the confidence interval means and the fact that a smaller interval with the same "confidence" means either our sample size or variability is smaller that, for either scenario, implies better data. Moreover, an efficient estimator requires fewer input data or observations (i.e. *n* is smaller) than a less efficient one, which is important when we explore the **Cramér–Rao bound**.

From 231, $\bar{x} \pm t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$